

Question 1

Question ID: 1587896

Lurry Corporation executives are considering the purchase of a factory that produces after-market parts for classic cars. The purchase price is \$38 million. They have estimated the after-tax cash flows from operating the factory to be \$3 million at the end of Year 1, \$4 million at the end of Years 2 and 3, and \$5 million at the end of Years 4 and 5. After five years, they expect to be able to sell the factory for \$40 million. Based on the company's 9.5% estimate of its weighted average cost of capital, the net present value (NPV) for the factory is:

A) \$0.98 million.



B) \$3.18 million.



C) \$4.12 million.



Explanation

The cash flows (in millions) are –38 at Year 0, 3 at Year 1, 4 at Years 2 and 3, 5 at Year 4, and 45 at Year 5. The NPV of the factory purchase, using a 9.5% discount rate, is \$3.1855 million.

(Module 26.1, LOS 26.b)

Question 2

Question ID: 1587863

Using a money-weighted return to evaluate an investment manager's performance is appropriate if the manager:

A) controls the cash flows into and out of the portfolio.



B) holds investments denominated in multiple currencies.



C) actively trades with an average holding period less than one year.



Explanation

If a manager controls the cash flows into and out of the portfolio, a money-weighted return is appropriate for evaluating the manager's performance. Otherwise, a time-weighted return is more appropriate.

(Module 1.2, LOS 1.c)

Question 3

Question ID: 1587864

An investor holds a bond that matures one year from today. When the bond matures, he will take the proceeds and purchase 1,000 shares of XYZ stock. If he assumes constant dividend growth, which of the following values will he need to estimate to determine the intrinsic value of the stock upon purchase?

A) The value of the bond on the maturity date.



B) The dividend expected to be paid two years from today.



C) A required return that must be less than the growth rate.



Explanation

The constant growth (Gordon growth) dividend discount model is used to calculate the intrinsic value of a stock. This value is a by-product of the dividend in the next period, the required return on common equity, and the constant growth rate of dividends. To determine the value one year from today, the investor will need to determine the dividend two years from today (the dividend used in the equation must always be one year beyond the date of the valuation), the required return (which must exceed the growth rate), and the growth rate itself. The value of the bond on the maturity date is presumed to be its face value.

(Module 2.1, LOS 2.a)

Question 4

Question ID: 1587887

The one-year interest rate in Minova (MNO) is 6% and the one-year interest rate in Piquar (PQR) is 8%. If the spot MNO/PQR exchange rate is 3.4500, the no-arbitrage one-year forward rate is *closest* to:

A) 3.3860.



B) 3.5150.



C) 3.6440.



Explanation

The no-arbitrage relationship may be stated as:

forward MNO/PQR = spot MNO/PQR ×

= 3.45 × (1.06 / 1.08) = 3.3861.

Notice this question can be answered without performing the calculation. Because the MNO interest rate is less than the PQR interest rate, the no-arbitrage forward MNO/PQR exchange rate must be less than the spot MNO/PQR, and only one of the answer choices is less than 3.4500.

(Module 2.2, LOS 2.c) and (Module 19.1, LOS 19.b)

Question 5

Question ID: 1587868

Raymond Green has created a pricing model for estimating future returns on high volume, large capitalization stocks. To keep the model relatively simple, Green has assumed that all returns are both identically and independently distributed. The assumptions of Green's model implies that:

A) stationarity is not factored into the model.



B) previous returns are helpful in forecasting future returns.



C) the mean and variance of returns are constant over time.



Explanation

For returns to be identically distributed, the mean and variance of returns are assumed to be constant over time (which is called stationarity). Independent distribution implies that past returns are *not* useful for predicting future returns.

(Module 6.1, LOS 6.a)

Question 6

Question ID: 1587871

Using a sample size of 58, an analyst determines that a sample correlation coefficient for two random variables is 0.21. The hypothesis at a 10% significance level is that the population coefficient is equal to zero. If the test statistic is 1.571 and the critical value is 1.6725, the analyst should:

A) fail to reject the null hypothesis.



B) reject the null hypothesis and determine the populations are correlated.



C) reject the null hypothesis and determine the populations are not correlated.



Explanation

Because the test statistic is below the critical value, the analyst cannot reject the null hypothesis that the population coefficient is equal to zero.

(Module 9.1, LOS 9.a)

Question 7

Question ID: 1587874

A data scientist who is engaging in curation is *most likely*:

A) collecting data.



B) archiving and accessing data.



C) adjusting for bad or missing data.



Explanation

Curation refers to assuring data quality, such as by adjusting for bad or missing data. Collecting data is referred to as capture. Archiving and accessing data is referred to as storage.

(Module 11.1, LOS 11.b)

Question 8

Question ID: 1587877

In the first two quarters of the calendar year, the GDP growth rate was -0.5% and -0.8% . In the most recently completed third quarter, the GDP growth rate was 1.5% . During that same third quarter, the inflation rate decreased from 6.3% to 5.5% while unemployment remained high but relatively stable. The business cycle phase during the third quarter is *best* characterized as a:

A) peak.



B) trough.



C) contraction.



Explanation

During the trough phase of the business cycle, the GDP growth rate changes from negative to positive (which it did in the third quarter here), unemployment is high, spending on housing and goods increases, and inflation may be moderate or decreasing (which it is here). A peak will reflect a decrease in GDP, low unemployment, and an increase in inflation. Contraction typically reflects negative growth GDP, increases in unemployment, and decreases in inflation.

(Module 13.1, LOS 13.a)

Question 9

Question ID: 1587878

An approaching peak in a business cycle is *most likely* indicated by an increasing:

A) GDP growth rate.



B) unemployment rate.



C) inventory/sales ratio.



Explanation

An unintended increase in inventories relative to sales is often an indicator that sales are slowing as an economic expansion approaches its peak. The rate of GDP growth, although positive, is typically decreasing in this stage of the business cycle. The unemployment rate typically does not begin increasing until after a contraction has begun because companies usually hesitate to dismiss workers until they are convinced a contraction will persist.

(Module 13.1, LOS 13.c)

Question 10

Question ID: 1587879

A government's budget deficit increases as a result of its legislature enacting an increase in spending. This action may be classified as a tool of:

A) fiscal policy only.



B) monetary policy only.



C) either fiscal or monetary policy.



Explanation

Changes in a government budget deficit or surplus represent fiscal policy. Monetary policy refers to the quantity of money and the amount of credit available in the economy and is typically determined by the country's central bank.

(Module 14.1, LOS 14.a)

Question 11

Question ID: 1587881

Due to a series of interest rate increases over the last couple of years designed to slow inflation, a country's economy has moved into a recession. The country's central bank has begun to take steps to reverse that trend through monetary policy tools. Which of the following is *most likely* to result from the central bank's recent measures?

A) An increase in the policy rate.



B) Appreciation of the domestic currency.



C) A decrease in short and long-term interest rates.



Explanation

To help move the economy out of a recession, the central bank will implement expansionary monetary policy. This will likely result in a decrease in the policy rate, a decrease in both short and long-term interest rates, and a depreciation of the domestic currency in the foreign exchange market.

(Module 15.1, LOS 15.b)

Question 12

Question ID: 1587884

Three countries enter into a regional trading agreement under which they establish an agreed-upon economic policy and adopt a single currency. This type of agreement is *best* described as a(n):

A) customs union.



B) economic union.



C) monetary union.



Explanation

In addition to removing import barriers, export barriers, and movement of labor and capital goods barriers, a monetary union establishes common institutions and economic policy for the union and utilizes a single currency for all member countries. Customs unions do not have common economic policies and do not adopt a single currency. An economic union is similar to a monetary union except that it does not use a single currency for all of its member countries.

(Module 17.1, LOS 17.c)

Question 13

Question ID: 1587885

In a foreign exchange transaction, the "buy side" refers:

A) only to the party acquiring the base currency.



B) only to the party acquiring the price currency.



C) to both the party acquiring the base currency and the party acquiring the price currency.



Explanation

In the foreign exchange market, the "buy side" refers to the parties that need to exchange currencies, and the "sell side" refers to the financial intermediaries through which these parties carry out foreign exchange transactions.

(Module 18.1, LOS 18.a)

Question 14

Question ID: 1587889

Investors who are concerned about potential exposure to an entity's liabilities should *least* appropriately participate in:

A) limited partnerships.



B) general partnerships.



C) ownership of publicly traded equities.



Explanation

General partnerships involve unlimited liability for owners. Limited partners have limited liability, as do equity investors in corporations.

(Module 22.1, LOS 22.a)

Question 15

Question ID: 1587890

In jurisdictions where double taxation of dividends occurs, shareholders' effective tax rate on dividends is:

A) unrelated to a company's dividend payout ratio.



B) less, the higher a company's dividend payout ratio.



C) greater, the higher a company's dividend payout ratio.



Explanation

The higher a company's dividend payout ratio is, the greater the shareholder's effective tax rate on dividends if the shareholder is subject to double taxation of dividends.

(Module 22.1, LOS 22.b)

Question 16

Question ID: 1587891

Which of the following statements about the differences between debt and equity investments is *most* accurate?

A) While equity holders can lose their entire investment, debtholders are protected from losing their principal.



B) Debt carries more risk than equity, which makes it costlier for the issuing entity to issue debt.



C) The upside potential for equity is greater because debtholders will not receive more than their interest and principal payments.



Explanation

For most debt investments, debtholders at best receive their scheduled interest payments on time and their principal investment paid back at maturity. Equity investors, by contrast, have theoretically unlimited upside potential. Debtholders can lose their principal if the issuer defaults on its obligations. Debt carries less risk than equity, which makes it less costly to the issuing entity because debt investors demand a lower required return than equity investors.

(Module 23.1, LOS 23.a)

Question 17

Question ID: 1587892

A principal-agent conflict *most likely* exists when a company's managers:

A) issue debt and use the proceeds to fund an increase in common dividends.



B) pursue unnecessary acquisitions because their compensation is based on the size of the company.



C) have effective control of the company because they hold shares with greater voting rights than other shareholders.



Explanation

"Empire building" that does not create value for shareholders, when it results from a compensation structure that rewards such behavior, is an example of a principal-agent conflict. Issuing debt to fund increased dividends represents a conflict between creditors and shareholders. Dual-class share structures are an example of conflicts among groups of shareholders.

(Module 24.1, LOS 24.a)

Question 18

Question ID: 1587893

A company's Board of Directors is *most likely* to assign responsibility for assessing the company's internal controls to the:

A) audit committee.



B) governance committee.



C) remuneration committee.



Explanation

Responsibility for the internal audit function and the effectiveness of a company's internal controls typically rests with the audit committee.

(Module 24.1, LOS 24.b)

Question 19

Question ID: 1587894

An analyst has discovered that over the last three years, Gathers Company has experienced a decrease in its net operating cycle, while over the same time period the average net operating cycle for the industry (excluding Gathers) has increased. These trends *most likely* indicate that:

A) Gathers has decreased its liquidity position by increasing the amount of time inventory spends in its warehouses.



B) Gathers has increased its liquidity position by increasing the speed of cash collection from its customers.



C) the industry has decreased its liquidity position by increasing the average amount of time to pay suppliers.



Explanation

The net operating cycle (also called the cash conversion cycle) measures the amount of time between paying the firm's suppliers for raw materials and collecting cash through the sale of finished goods.

$$\text{Net Operating Cycle} = \text{number of days of inventory} + \text{number of days of receivables} - \text{number of days of payables}$$

Gathers Company has decreased its net operating cycle, indicating it has decreased the number of days of inventory (more efficiently managed inventory purchases, processing, and fulfillment), decreased the number of days of receivables (reduced credit terms to customers and reduced collection times), or increased the number of days of payables (lengthened the amount of time to pay suppliers). The industry, however, has increased its average net operating cycle over the same period, indicating the industry has become less liquid by increasing number of days of inventory or number of days of receivables or decreasing number of days of payables.

Question 20

Question ID: 1587895

A company with a more conservative approach to liquidity management relative to its peers is *most likely* to have higher:

A) profitability.



B) current assets as a percentage of total assets.



C) short-term debt as a percentage of total debt.



Explanation

A conservative approach to managing liquidity is to hold relatively high levels of current assets and finance working capital with equity and long-term debt, but this approach may decrease a company's profitability. Financing working capital with short-term debt is a more aggressive approach.

(Module 25.1, LOS 25.c)

Question 21

Question ID: 1587897

QuarryCo is determining whether to expand its current production capacity. A feasibility study completed one year ago indicates that the rock in the new quarry site is of sufficient quality. The project would require an increase in working capital and the use of an empty factory owned by the company. Several existing customers would be expected to purchase materials from the new quarry due to its closer proximity. In evaluating the expansion project, QuarryCo should *least appropriately* consider:

A) cash expended to perform the feasibility study.



B) the increase in working capital required to support the project.



C) the effects of customers who will switch purchases to the new quarry.



Explanation

Cash previously spent to perform a feasibility study is a sunk cost that should be ignored. Working capital requirements and cannibalization are factors that should be considered in capital budgeting.

(Module 26.2, LOS 26.c)

Question 22

Question ID: 1587898

Project 1 has a positive NPV. Project 2 is identical to Project 1, except that Project 2 can be delayed by a year if market conditions are unfavorable. An analyst should most likely consider Project 2 to have:

A) less value than Project 1.



B) the same value as Project 1.



C) greater value than Project 1.



Explanation

Project 2 includes a timing option. An analyst should add an estimated value of this real option to the project's NPV.

(Module 26.2, LOS 26.d)

Question 23

Question ID: 1587899

The following information refers to a company's capital structure and a marginal tax rate of 40%:

Type of Capital	Weight in Capital Structure	Before-Tax Component Cost
Bonds	40%	7.5%
Preferred stock	5%	11%
Common stock	55%	15%

The company's weighted average cost of capital is *closest* to:

A) 10.0%.



B) 10.6%.



C) 11.8%.



Explanation

$$WACC = (w_d)(k_d)(1 - t) + (w_{ps})(k_{ps}) + (w_{ce})(k_{ce}) = (0.4)(7.5)(1 - 0.4) + (0.05)(11) + (0.55)(15) = 10.6\%$$

Question 24

Question ID: 1587900

Which set of assumptions is *most likely* consistent with a company's optimal capital structure being 100% debt?

A) No taxes and no costs of financial distress exist.



B) No costs of financial distress exist, but taxes do exist.



C) Both taxes and costs of financial distress exist.



Explanation

The results of Modigliani and Miller's capital structure model with taxes, but with no costs of financial distress, are that firm value would be maximized by financing 100% with debt.

(Module 27.2, LOS 27.c)

Question 25

Question ID: 1587901

Which of the following recommendations *most likely* represents an element of a company's channel strategy?

A) The ideal customers are ages 18 – 30 in the western United States.



B) Focus the sales of products through online and direct mail distribution mechanisms.



C) Given the competition and the breadth of substitute products, value-based pricing should work well.



Explanation

A channel strategy specifies how a firm will sell its products or services to its customers. Options for getting products to customers include direct sales or through intermediaries, as well as online, direct mail, trade shows, and physical locations. Identifying the firm's potential customers is called segmenting, which is an aspect of a company's business model. Pricing strategy is independent of a channel strategy.

(Module 28.1, LOS 28.a)

Question 26

Question ID: 1587865

Returns data for Limbo Company exhibit the following statistics:

Mean	9.5%
Median	14.3%
Excess Kurtosis	-0.97

The returns distribution for Limbo Company is:

A) positively skewed.



B) negatively skewed.



C) not skewed.



Explanation

Since the median is higher than the mean, the distribution is negatively skewed. If the mean were higher than the median the distribution would be positively skewed.

(Module 3.2, LOS 3.c)

Question 27

Question ID: 1587866

Cheryl Smith, CFA, is comparing dividend changes for energy and non-energy companies. Smith determines that 15% of the stock market universe consists of energy companies. Smith also determines that the probability that an energy company will increase its dividend is 90% and the probability that a non-energy company will increase its dividend is 30%. If Smith randomly selects one company from the universe of stocks and notices that the company declared a dividend increase, the probability that the company Smith selected is an energy company is *closest* to:

A) 5%.



B) 15%.



C) 35%.



Explanation

Initially, we are told that the probability of randomly selecting an energy company is 15% ("Smith determines that 15% of the stock market universe consists of energy companies"). This is the prior probability. But Smith gathers new information that she can use to update her prior probability—the randomly selected stock recently declared a dividend increase. Intuitively, we know the updated probability will be much higher than 15% because dividend increases are likely to occur much more often for energy companies (90% of the time) than for non-energy companies (30% of the time). Thus, the correct choice must be 35%.

We can obtain the exact answer using Bayes' formula:

$$\text{updated probability} = P(E|D) = P(E) \times \frac{P(D|E)}{P(D)}$$

where:

- $P(E)$ is the prior probability that the selected company is an energy company.
- $P(D|E)$ is the probability of the new information (dividend increase) for a given event (energy company).
- $P(D)$ is the unconditional probability of the new information (dividend increase).

$P(E)$ is provided in the question as 15%. $P(D|E)$ also is provided in the question as 90%. $P(D)$ is the probability of a dividend increase, which can occur either if the randomly selected company is an energy company or a non-energy company:

$$P(D) = [P(E) \times P(D|E)] + [P(E^c) \times P(D|E^c)] = [0.15 \times 0.90] + [0.85 \times 0.30] = 0.39$$

Therefore, updated probability = $0.15 \times \frac{0.90}{0.39} = 0.346 = 34.6\%$.

(Module 4.1, LOS 4.c)

Question 28

Question ID: 1587867

Given the information in the table for Portfolios A, B, and C, and a target return of 3%, which portfolio has the lowest shortfall risk?

Portfolio	A	B	C
Expected Return	13%	9%	15%
Standard Deviation	5.1%	3.6%	7.2%

A) Portfolio A.



B) Portfolio B.



C) Portfolio C.



Explanation

The portfolio with the lowest probability of returns less than 3% is the one with the highest safety-first ratio.

Safety-first ratio Portfolio A: $(13 - 3) / 5.1 = 1.96$

Safety-first ratio Portfolio B: $(9 - 3) / 3.6 = 1.67$

Safety-first ratio Portfolio C: $(15 - 3) / 7.2 = 1.67$

Question 29

Question ID: 1587869

According to the Central Limit Theorem:

A) the distribution of sample means will be approximately normally distributed only if the population is normally distributed and continuous. 

B) inferences about the population mean can be made from the sample mean, as long as the sample size is sufficiently large. 

C) the sample mean will have a standard deviation equal to the population standard deviation divided by the sample size. 

Explanation

According to the Central Limit Theorem, if the sample size is large, the sample mean will be distributed normally regardless of the population's distribution, specific inferences can be about the population mean, and the sample mean will have a standard deviation equal to the population standard deviation divided by the square root of the sample size (also known as the standard error).

(Module 7.1, LOS 7.b)

Question 30

Question ID: 1587870

George Hutchins, CFA, would like to perform a paired comparisons test on returns for the stocks of two real estate investment trusts. The test statistic that Hutchins should select for the paired comparisons test is the:

A) t -statistic.



B) F -statistic.



C) Chi-square statistic.



Explanation

The paired comparisons test is performed using a t -statistic with $n - 1$ degrees of freedom.

(Module 8.2, LOS 8.b)

Question 31

Question ID: 1587872

In a simple linear regression, the regression line minimizes the:

A) total sum of squares.



B) sum of squared errors.



C) regression sum of squares.



Explanation

The regression line from a simple linear regression is the line that minimizes the sum of squared errors, which represents the variation in the dependent variable not explained by the linear relationship.

(Module 10.2, LOS 10.d)

Question 32

Question ID: 1587873

For which functional form of a linear regression is the slope coefficient interpreted as the absolute change in the dependent variable for a relative change in the independent variable?

A) Lin-log model.



B) Log-lin model.



C) Log-log model.



Explanation

With a lin-log model, we interpret the slope coefficient as the absolute change in the dependent variable for a relative change in the independent variable. Such a model is appropriate if the dependent variable is linear while the independent variable is logarithmic.

(Module 10.3, LOS 10.f)

Question 33

Question ID: 1587875

A firm under perfect competition finds that the market price of its product is below the firm's average total cost but above its average variable cost. If these conditions persist, the firm should:

A) shut down in the short run and go out of business in the long run.



B) shut down in the short run but continue operating in the long run.



C) continue operating in the short run but go out of business in the long run.



Explanation

Because the market price still allows the firm to recover all of its variable costs ($P > AVC$) and some of its fixed costs, it should continue operating in the short run and experience smaller losses compared to shutting down immediately. However, if the price is expected to remain below ATC, the firm should go out of business in the long run.

(Module 12.1, LOS 12.a)

Question 34

Question ID: 1587876

Assume a cartel is organized among the producers of a commodity and begins practicing collusion. The *most likely* effects on price and output are that:

A) both will increase.



B) price will increase and output will decrease.



C) price will decrease and output will increase.



Explanation

Collusion is an agreement among firms to avoid various competitive practices. The cartel practicing collusion will be similar to a monopoly, causing prices to increase and output to decrease compared to a competitive market.

(Module 12.2, LOS 12.d)

Question 35

Question ID: 1587882

Functions of a central bank *most likely* include:

A) collecting tax payments.



B) balancing the national budget.



C) controlling money supply growth.



Explanation

Central banks control the growth of a country's money supply through their monetary policy actions. Many central banks also regulate their countries' banking systems and issue their countries' currencies. Fiscal policy (taxation and government spending) is generally the responsibility of a country's executive and legislative officials.

(Module 15.1, LOS 15.a)

Question 36

Question ID: 1587880

Tax policy is said to exhibit efficiency if it:

A) is simple to implement and enforce.



B) causes minimal interference with market forces.



C) generates just enough revenues to fund government spending.



Explanation

Tax policy is said to exhibit efficiency if it minimizes its interference with market forces and work incentives.

(Module 14.2, LOS 14.c)

Question 37

Question ID: 1587883

Promoting stability in exchange rates is among the goals of the:

A) World Bank.



B) World Trade Organization.



C) International Monetary Fund.



Explanation

Promoting exchange rate stability is one of the stated goals of the IMF.

(Module 16.1, LOS 16.c)

Question 38

Question ID: 1587888

Assume the exchange rate between the Bucas (BCS) and the Leider (LDR) is 1.70 BCS/LDR, and the exchange rate between the Bucas and the Passoa (PAS) is 3.2 BCS/PAS. The PAS/LDR exchange rate is *closest* to:

A) 0.5313 PAS/LDR.



B) 1.8824 PAS/LDR.



C) 5.4400 PAS/LDR.



Explanation

The PAS/LDR cross rate is $(1.70 \text{ BCS/LDR}) / (3.2 \text{ BCS/PAS}) = 0.5313 \text{ PAS/LDR}$.

(Module 19.1, LOS 19.a)

Question 39

Question ID: 1587886

The type of exchange rate regime under which a country implements a monetary policy that is necessarily limited in scope by market foreign exchange rates is *most accurately* described as:

A) dollarization.



B) a monetary union.



C) a fixed peg arrangement.



Explanation

With a fixed peg arrangement, the country must pursue a monetary policy that keeps the exchange rate at the specified (peg) level, effectively adopting the monetary policy of the country issuing the currency to which its currency is pegged. There is no monetary policy with dollarization, as the country does not create money. With a monetary union, the country does not have an independent monetary policy, and the central bank for the union is not limited in its monetary policy.

(Module 18.2, LOS 18.b)

Question 40

Question ID: 1587902

Information about any conflicts of interest between management, the board of directors, and shareholders can *most likely* be found in the:

A) proxy statement.



B) footnotes.



C) auditor's report.



Explanation

The proxy statement provides information about management and board member compensation, as well as any conflicts of interest.

(Module 29.1, LOS 29.e)

Question 41

Question ID: 1587903

Pollard Construction has a contract with a customer to build a new factory over a period of three years. Total revenues from the project will be \$4 million. Pollard estimates the costs of construction to be \$800,000 in Year 1, \$600,000 in Year 2, and \$1.6 million in Year 3. Pollard expects to bill the customer for \$1 million in Year 1, \$1 million in Year 2, and \$2 million in Year 3. Profit recognized by Pollard in Year 2 of the project is:

A) zero.



B) \$200,000.



C) \$400,000.



Explanation

Pollard should use progress toward completion to account for long-term project revenue and profits. Total costs are \$3 million, total revenues are expected to be \$4 million, and total profit is the difference of \$1 million. Percentage of the total costs incurred in the second year is $600,000 / 3,000,000 = 20\%$. Profits recognized in the second year are $20\% \times 1 \text{ million} = \$200,000$.

(Module 30.1, LOS 30.a)

Question 42

Question ID: 1587904

At the beginning of the year, Weatherford Corporation had 2,000,000 shares of common stock outstanding. In addition, Weatherford had 150,000 stock options outstanding to purchase common shares at \$10 per share. No stock options were exercised during the year. Assuming the average market price of the stock was \$15, how many shares should Weatherford use in computing diluted earnings per share for the year?

A) 2,000,000.



B) 2,050,000.



C) 2,150,000.



Explanation

Diluted EPS is computed assuming conversion of the options using the treasury stock method. Accordingly, an additional 50,000 shares are added to compute diluted EPS: $(\$15 \text{ average market price} - \$10 \text{ exercise price}) / \$15 \text{ average market price} \times 150,000 \text{ options} = 50,000 \text{ shares}$. $2,000,000 \text{ shares} + 50,000 \text{ shares} = 2,050,000 \text{ shares}$.

(Module 30.4, LOS 30.d)

Question 43

Question ID: 1587905

U.S. GAAP requires unrealized gains and losses on available-for-sale securities to be reported:

A) in non-operating income.



B) in a separate section following net income.



C) as other comprehensive income on the statement of changes in owners' equity.



Explanation

Unrealized gains and losses on securities classified as available-for-sale are recorded as increases (gains) or decreases (losses) in other comprehensive income. Unrealized gains and losses on actively traded securities would be reported in the income statement.

(Module 31.1, LOS 31.c)

Question 44

Question ID: 1587906

A company's investments in marketable securities include a 3-year tax-exempt bond measured at amortized cost and a 5-year government note measured at fair value through other comprehensive income. On its income statement, the company should report the coupon interest received from:

A) both of these securities.



B) neither of these securities.



C) only one of these securities.



Explanation

Interest and dividends received are reported as income, regardless of the balance sheet classification of marketable securities.

(Module 31.1, LOS 31.c)

Question 45

Question ID: 1587908

Which of the following *most accurately* describes cash flow classification under U.S. GAAP and IFRS?

A) Dividends paid are a financing activity under U.S. GAAP and dividends received may be shown as an operating or investing activity under IFRS. 

B) Dividends received may be shown as an operating or investing activity under U.S. GAAP and dividends paid is a financing activity under IFRS. 

C) Interest expense is a financing activity under U.S. GAAP and interest received may be shown as an operating or investing activity under IFRS. 

Explanation

Interest received, dividends received, and interest paid are operating cash flows under U.S. GAAP, and dividends paid are financing cash flows. Interest received and dividends received may be shown as operating or investing cash flows under IFRS. Interest paid and dividends paid may be shown as operating or financing cash flows under IFRS.

(Module 32.3, LOS 32.d)

Question 46

Question ID: 1587909

An accountant with Umble Company is preparing the statement of cash flows. Cash flow from operations is \$210 and cash on the balance sheet increased by \$340. Transactions during the period include:

Capital expenditures	\$100
Investment in joint venture	\$40
Acquisitions	\$80
Dividends from affiliates	\$25

Umble's cash flow from financing (CFF) under U.S. GAAP is:

A) -\$220.



B) +\$195.



C) +\$350.



Explanation

CFI includes capital expenditures, the investment in a joint venture, and acquisitions: $-100 - 40 - 80 = -220$. The dividend from affiliates is included in operating cash flow. CFF is equal to total cash flow minus CFO minus CFI: $340 - 210 - (-220) = 350$.

Question 47

Question ID: 1587907

Selected year-end values for Argus Inc. are presented in the following tables:

Balance Sheet Items	December 31, 20X1	December 31, 20X2
---------------------	-------------------	-------------------

Accounts Receivable	\$20 million	\$19 million
Trade Payables	\$14 million	\$15 million
Inventory	\$27 million	\$28 million

Income Statement Items	December 31, 20X2
------------------------	-------------------

Net Income	\$30 million
Gain on Sale	\$1 million
Depreciation	\$1 million

Based only on the data in the tables, 20X2 cash flow for operations for Argus Inc. is:

A) \$27 million.



B) \$29 million.



C) \$31 million.



Explanation

Starting with net income of 30, add 1 for the decrease in receivables (a source of cash), add 1 for the increase in payables (a source of cash), subtract 1 for the increase in inventory (a use of cash), subtract 1 for the gain (non-CFO item), and add 1 for depreciation (noncash item), resulting in CFO = \$31 million.

(Module 32.1, LOS 32.b)

Question 48

Question ID: 1587920

Which of the following ratios can be read directly from a common-size income statement or balance sheet?

A) Return on assets.



B) Inventory turnover.



C) Operating profit margin.



Explanation

A common-size income statement reports items as a percentage of sales. The operating profit margin is operating earnings/sales so it can be read directly from a common-size income statement. Inventory turnover and return on assets divide an income statement item by a balance sheet item.

(Module 39.1, LOS 39.a)

Question 49

Question ID: 1587921

At the end of last year, Manhattan Corporation had a quick ratio of 1.2. If Manhattan reduces its accounts payable with a cash payment of \$2 million, its quick ratio will:

A) be unchanged.



B) increase.



C) decrease.



Explanation

Since the quick ratio is greater than one, the percentage decrease in the denominator (current liabilities) is greater than the percentage decrease in the numerator (cash + accounts receivable + marketable securities). The denominator will decrease relatively more than the numerator. As a result, the quick ratio will increase.

(Module 39.2, LOS 39.b)

Question 50

Question ID: 1587922

An analyst is *most likely* to be concerned about the predictability of a firm's sales if, compared to its industry competitors, the firm has a significantly higher:

A) standard deviation of sales.



B) coefficient of variation of sales.



C) correlation of sales with economic growth.



Explanation

Coefficient of variation of sales (standard deviation of sales / mean sales) is a measure analysts can use to compare the sales variability of different firms. Standard deviation of sales by itself would be expected to be higher for firms with greater sales revenue. High correlation of sales with economic growth suggests a firm is cyclical and therefore expectations about its sales can be based on business cycle expectations.

(Module 39.5, LOS 39.e)

Question 51

Question ID: 1587910

A firm that reports under IFRS wrote down its inventory from cost of \$240,000 to net realizable value of \$210,000. In the next period, cost was unchanged, but net realizable value (NRV) increased to \$250,000. The firm will *most appropriately* report ending inventory for the period as:

A) \$210,000.



B) \$240,000.



C) \$250,000.



Explanation

Under IFRS, inventory can be written up when NRV increases but cannot be carried at an amount greater than cost. In this case, the firm can reverse the previously recognized write-down, but it may not write up inventory above its cost of \$240,000.

(Module 34.1, LOS 34.a)

Question 52

Question ID: 1587911

Compared to capitalizing a purchase, expensing a purchase will result in:

A) higher operating cash flow and lower investing cash flow.



B) higher operating cash flow and higher investing cash flow.



C) lower operating cash flow and higher investing cash flow.



Explanation

Expensing the purchase will decrease CFO and increase CFI because the expenditure will be treated as an operating cash outflow rather than an investing cash outflow.

(Module 35.1, LOS 35.a)

Question 53

Question ID: 1587912

The present value of an equity security's future cash flows is *most likely* to be significantly different from the company's:

A) book value per share.



B) market value per share.



C) intrinsic value per share.



Explanation

A company's book value per share is typically not equal to its intrinsic value per share, since book value is based on the historical cost of assets. Intrinsic value is the present value of the security's future cash flows. The market value of a share is the cumulative investors' estimation of this intrinsic value, and market value is usually somewhat in line with the intrinsic value.

(Module 44.2, LOS 44.g)

Question 54

Question ID: 1587913

How is the impact of selling a long-lived asset recorded on a firm's income statement?

A) The sale proceeds increase revenue and the carrying value increases cost of goods sold.



B) The difference between the sale proceeds and the carrying value is reported as a gain or loss.



C) The difference between the sale proceeds and the original value is reported as a gain or loss.



Explanation

The difference between the sale proceeds and the carrying value for a long-lived asset is reported on a firm's income statement as a gain or loss. The carrying value is the original value less accumulated depreciation.

(Module 35.2, LOS 35.b)

Question 55

Question ID: 1587916

A financial analyst should treat deferred tax liabilities (DTLs) as equity if the DTLs are:

A) due to permanent differences.



B) expected to increase each period.



C) expected to decrease each period.



Explanation

If a firm's deferred tax liabilities are expected to increase each period, they are not expected to reverse and therefore the deferred taxes are not expected to be paid in the foreseeable future. For analysis, the deferred tax liability can be removed and the decrease in liabilities will increase equity. Permanent differences do not create deferred tax items.

(Module 37.2, LOS 37.b)

Question 56

Question ID: 1587915

Consider a firm that has both deferred tax assets and deferred tax liabilities on its balance sheet. What will be the effect on the deferred tax asset and liability of an increase in the firm's marginal tax rate?

A) Both will increase.



B) Both will decrease.



C) One will increase and one will decrease.



Explanation

Under the liability method, both deferred tax assets and deferred tax liabilities will increase when the tax rate increases because the future tax rate is assumed to increase.

(Module 37.1, LOS 37.a)

Question 57

Question ID: 1587917

Manitou Plastics, Inc., has been recording large deferred tax assets after incurring operating losses in the previous three years. At the end of the most recent year, Manitou reported \$14 million in deferred tax assets but only \$3 million in deferred tax liabilities. Manitou has also reported a valuation allowance related to deferred taxes in the amount of \$7.5 million. What is the *most likely* cause of Manitou's reported valuation allowance?

A) Accounting earnings have been manipulated.



B) Future profitability is in doubt.



C) Interest rates have increased.



Explanation

Valuation allowances occur when the probability of utilizing deferred tax assets is in doubt. Deferred tax assets must be reduced by a valuation allowance to reflect the probability that they will never be used.

(Module 37.2, LOS 37.b)

Question 58

Question ID: 1587914

For a lessor, how are lease payments treated on the cash flow statement?

A) The entire lease payment is an investing cash inflow.



B) The entire lease payment is an operating cash inflow.



C) The principal portion is an investing cash inflow, and the interest portion is an operating cash inflow.



Explanation

For a lessor, lease payments are classified as operating cash inflows under both IFRS and U.S. GAAP.

(Module 36.1, LOS 36.a)

Question 59

Question ID: 1587918

Which of the following choices about accounting for property, plant, and equipment are *most likely* to reflect a conservative bias?

A) Accelerated depreciation, longer useful lives, and lower salvage values.



B) Straight-line depreciation, longer useful lives, and higher salvage values.



C) Accelerated depreciation, shorter useful lives, and lower salvage values.



Explanation

Conservative accounting recognizes lower earnings in the current period and higher earnings in later periods. With respect to property, plant, and equipment, management with a conservative accounting bias would recognize the most depreciation expense (and thus the lowest earnings) in the current period by choosing an accelerated depreciation method and estimating shorter useful lives and lower salvage values.

(Module 38.1, LOS 38.c)

Question 60

Question ID: 1587919

Other things equal, what impact will increasing days sales in payables have on operating cash flow?

A) No impact.



B) Lower operating cash flow.



C) Higher operating cash flow.



Explanation

Stretching accounts payable (increasing days sales in payables) will increase operating cash flow.

(Module 38.3, LOS 38.h)

Question 61

Question ID: 1587982

If an investment ends the week above where it started the week, the investor will gain \$3,000. If the investment ends the week at or below where it started the week, the investor will earn \$0. The investment has a 60 percent probability of ending the week higher. With regard to this opportunity:

A) a risk-averse investor will prefer a certain payment of \$1,800 to this investment.



B) a risk-neutral investor will prefer this investment to a certain payment of \$1,800.



C) a risk-seeking investor will be indifferent between a certain payment of \$1,800 or this investment.



Explanation

The expected payoff is $\$3,000 \times 60\% + \$0 \times 40\% = \$1,800$. A risk-averse investor would prefer a certain payoff of this amount over taking the chance of a \$0 payout. A risk-neutral investor would be indifferent between a certain payout of \$1,800 or an expected value of \$1,800. A risk-seeking investor would prefer the uncertain outcome with a potentially greater upside to a certain payout of \$1,800.

(Module 20.2, LOS 20.b)

Question 62

Question ID: 1587983

A portfolio is 45% invested in stocks that have a standard deviation of returns of 27%, and 55% invested in bonds that have a standard deviation of returns of 14%. The covariance of stock and bond returns is 0.02. The standard deviation of the portfolio returns is *closest* to:

A) 3.0%.



B) 14.5%.



C) 17.5%.



Explanation

$$\sigma_{\text{portfolio}} = \sqrt{(0.45^2)(0.27^2) + (0.55^2)(0.14^2) + 2(0.45)(0.55)(0.02)} = 17.49\% = 17.49\%$$

(Module 20.3, LOS 20.e)

Question 63

Question ID: 1587984

Historically, returns on major asset classes have exhibited:

A) positive skewness and negative excess kurtosis.



B) negative skewness and positive excess kurtosis.



C) negative skewness and negative excess kurtosis.



Explanation

Returns on major asset classes have exhibited negative skewness (which reflects larger and more frequent negative deviations from their mean returns than a normal distribution) and positive excess kurtosis (which reflects larger and more frequent positive and negative deviations from their mean returns than a normal distribution).

(Module 20.3, LOS 20.d)

Question 64

Question ID: 1587985

The curve representing the set of portfolios that has the highest expected return for a given level of risk is the:

A) utility curve.



B) efficient frontier.



C) indifference curve.



Explanation

The efficient frontier represents the set of portfolios that has the highest expected return for a given level of risk. An indifference curve (in modern portfolio theory) represents the risk and return combinations that yield the same level of utility for an investor. An investor's utility curve (in modern portfolio theory) measures an investor's utility as a function of risk and return.

(Module 20.4, LOS 20.g)

Question 65

Question ID: 1587986

Which of the following statements about the capital market line (CML) is *most accurate*?

A) Only risky portfolios plot on the CML.



B) Only efficient portfolios plot on the CML.



C) In equilibrium, all portfolios plot on the CML.



Explanation

The capital market line is the efficient frontier when a risk-free asset is combined with the market portfolio. Therefore, only efficient portfolios plot on the CML. A portfolio that contains only the risk-free asset plots on the CML at its intercept with the vertical axis.

(Module 21.1, LOS 21.b)

Question 66

Question ID: 1587987

Based on the advice of his financial advisor, an investor sells a stock that makes up a large portion of his portfolio and reinvests those funds into 20 unique stocks with relatively low and negative correlations of returns. These actions will have the *most* direct impact on which component of portfolio risk?

A) Market risk.



B) Systematic risk.



C) Nonsystematic risk.



Explanation

Nonsystematic risk (also known as unsystematic risk, unique risk, diversifiable risk, or firm-specific risk) is the risk that can be reduced or even eliminated via a well-diversified portfolio. Investing in multiple securities with low or negative correlations can decrease exposure to nonsystematic risk. Systematic risk (also known as market risk or non-diversifiable risk) cannot be diversified away regardless of the number of investments in a portfolio.

(Module 21.1, LOS 21.c)

Question 67

Question ID: 1587988

Carla Duckett develops a return generating model that estimates expected returns on stocks based on their factor sensitivities to long-term interest rates and the growth rate of operating cash flow. Duckett's model is *best* described as using:

A) statistical and fundamental factors.



B) macroeconomic and statistical factors.



C) fundamental and macroeconomic factors.



Explanation

Long-term interest rates are a macroeconomic factor, and the growth rate of operating cash flow is a fundamental factor. Statistical factors are variables other than macroeconomic or fundamental factors that exhibit a statistical relationship with security returns.

(Module 21.1, LOS 21.d)

Question 68

Question ID: 1587989

Two stocks, Shaw Inc., and Melon Inc., have identical total risk. The Shaw stock risk is composed of 60% systematic risk and 40% unsystematic risk, while the Melon stock risk is composed of 40% systematic risk and 60% unsystematic risk. In equilibrium, according to capital market theory, Shaw has:

A) a higher expected return than Melon.



B) a lower expected return than Melon.



C) the same expected return as Melon.



Explanation

Both stocks have the same total risk, but Shaw has more systematic risk (higher beta) than Melon. In equilibrium, both stocks will plot on the Security Market Line, and the expected return will be greater for the higher-beta stock.

(Module 21.2, LOS 21.h)